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Mohssin Ibna Hossain
AIUB. Accounting Notes.

Final Term

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Ch 10:

Plant Assets, Natural Resources, And Intangible Assets

GOOD LUCK™

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#Plant Asset Expenditure:—

⇒ Plant assets are resource that have.

① Physical substance.

② Are used in the operation of a business.

③ are not intended for sale to customers.

④ are expected to be of use to the company for a Number of years.

Example:—

(i) property, (ii) plant, (iii) equipment. (iv) Fixed Assets.

#Determine the Cost of Plant Assets:—

Cost typically include:—

① Cash purchase price.

② title and Attorney's fees.

③ real estate broker's commissions.

④ property taxes and others.

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problem:— GP company purchases a delivary truck at a cash price 22,000 tk, Related expenditure are sales taxes 13,20 tk, painting and lettering 500 tk, motor licence 80 tk, three years insurance 1600 tk. (i) Compute the cost of the delevary truck (ii) Prepare the journal entry

⇒

(i)

	Delivery truck
Cash price	Tk 22,000
Sales taxes	1320
Painting and Lettering	500
Cost of delivery truck	Tk. 23,820

(ii)

	Dr	Cr
Equipment	23,820	
License Expense	80	
prepait Insurance	1600	
Cash		25,500

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Depreciation Method:—

⇒ Management selects the method it believes best measures an assets contribution to revenue over it's useful life.

there are three types of method.

1. Straight - line method.
2. Units - of - activity method.
3. Declining - balance method.

Straight - line method:—

Depreciation Expense = [Cost - Salavage value] / useful Life

Depreciable Cost = Cost - Salavage Value

∴ Depreciation Expense = Dipreciable cost / useful Life OR

Depreciation Expense = Depreciable cost × Straight Line Rate

Cost = purchase price + import duty + VAT

Book value = Cost - Accumulated Depreciation here,

Cost = নির্মাণ ব্যয়

Salvage value = অবশিষ্ট টুকরার মূল্য

$$\boxed{\text{Cost} = \text{price} + \text{import duty} + \text{VAT}}$$

Note:- Insurance related expense, shipping, set up cost, training or testing cost, এই সবগুলোই হবে Assets.

#Units-of-Activity Method:-

$$\boxed{\text{Depreciation Expense} = \text{Actual usage or Output units} \times \text{Depreciable cost per Unit}}$$

$$\boxed{\text{Depreciable cost per Unit} = [\text{cost} - \text{Salvage Value}] / \text{Total Units}}$$

here,

Depreciable cost per unit = প্রতি unit এ cost.

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Double-Declining-Balance Method:-

$\boxed{\text{Depreciation Expense} = \text{Beg. NBV} \times \text{D.B. Rate}}$

Beg = Begining of financial period.

NBV = Net Book Value = Cost - Accumulated Depreciation

D.B. Rate = Declining balance rate = proxy in double straight-line rate

problem:-

⇒ R-L-C Company purched a machine for there factory, the price of the machine 50,00,000 taka, Import duty 50,000 taka shipping and transportation cost 1,00,000 taka. purched oil/Grease for the meachine 20,000 taka, Installation or setup cost 50,000 taka.

(i) Find the cost of the purched machine.

(ii) prepare the journal Entry

(i)

price	TK 50,00,000
Import duty	50,000
Shipping/transportation	1,00,000
Intatall/setup	50,000
Cont of the purchad machine	TK 52,00,000

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	Dr	Cr
Equipment	52,00,000	
Supplies	20,000	
Cash		52,20,000

#Problem1:

➔Barb's Florists purchased a small delivery truck at a cost of $13,000 on January 1, 2020. DuPage expects the truck to have a salvage value of $1,000 at the end of its 5-year useful life.

During its useful life, the truck is expected to be cover 100,000 miles. Actual annual miles cover was 2020, 15,000; 2021, 30,000; 2022, 20,000; and 2023, 25,000 and 2024, 10,000

Required: Compute depreciation using the following.

(a) Straight-Line (b) Units-of-Activity (c) Declining Balance

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(i)

We know:-

Depreciable cost = Cost - sal vage value
= $13,000 - $1 000
= $12,000

Depreciation expanse = Depreciable cost ÷ useful life.
= $12,000 ÷ 5
= $ 2400

Depreciation Rate = $\frac{2400}{12000} \times 100\%$

= 20%

Straight line method

Computation

Year	Depreciable Cost	× Depreciation Rate	= Anual Depreciation Expense	End of year Accumulated Depreciation	Book value
2020	$12,000	20%	$2400	$24,00	$10,600
2021	12,000	20%	2400	4800	8200
2022	12,000	20%	2400	7200	5800
2023	12,000	20%	2400	9600	3400
2025	12000	20%	2400	12000	1000

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We know:—

Depreciable cost per unit = (cost − salvage value) ÷ total Units

= ($13000 − $1000) ÷ 10,0000

= $0.12

∴ Units of Activity Method:—

Barb's Florists:—

Year	Computation				End of year	
	Unit of Activity	× Depreciable Cost / Unit	= Depreciation Expense		Accumulated Depreciation	Book value
2020	15,000	$ 0.12	$1800		$ 1800	$11200
2021	30,000	0.12	3600		5400	7600
2022	20,000	0.12	2400		7800	5200
2023	25,000	0.12	3000		10800	2200
2024	10,000	0.12	1200		12000	1000

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(iii)

We know:-

D.B = porary in double straight line rate
rate

$\therefore$ D.B. Rate = 2 × 20% = 40%.

Declining - Balance - Method

Barb's Florists

	Computation.			End of year	
year	Book value Beg. of year ×	Depreciation Rate =	Annual Depreciation Expense	Accumulated Depreciation	Book value
2020	$ 13,000	40%	$ 5200	$ 5200	$ 7800
2021	7800	40%	3120	8320	4680
2022	4680	40%	1872	10192	2808
2023	2808	40%	1123	11315	1685
2024	1685	40%	685	12000	1000

Computation of $674 ($1685 × 40%) in adjusted to $685 for book value to equal salvage value.

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#Problem 2:

Barb's Florists purchased a small delivery truck at a cost of $13,000 on **April 1, 2020**. DuPage expects the truck to have a salvage value of $1,000 at the end of its 5-year useful life.

During its useful life, the truck is expected to be cover 100,000 miles. Actual annual miles cover was 2020, 10,000; 2021, 30,000; 2022, 20,000; 2023, 25,000 and 2024, 10,000 and 2025, 5,000

Required: Compute depreciation using the following.

(a) Straight-Line (b) Units-of-Activity (c) Declining Balance

(a)

We know—

Depreciable Cost = Cost − salvage value

$= \$13000 - \$1000$

$= \$12000$

Depreciation expense = Depreciable cost / useful life.

$= \$12000 \div 5$

$= \$2400$

Barb's Florists

Year	Computation: Depreciable Cost	× Rate	= Annual Depreciation Expense	× Partial Year	= Current Year Expense	End of year: Accumulated Depreciation	End of year: Book Value
2020	\$12000	× 20%	= \$2400	× $\frac{9}{12}$	= \$1,800	\$1800	11200
2021	12000	20%	2400		2400	4200	8800
2022	12000	20%	2400		2400	6600	6400
2023	12000	20%	2400		2400	9000	4000
2024	12000	20%	2400		2400	11400	1600
2025	12000	20%	2400	× $\frac{3}{12}$	= 600	12000	1000
					\$12,000		

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(b)

We know:—

Depreciable cost per unit = Depreciable cost / total units
= (Cost - salvage value) / total units.
= ($13000 - $1000) ÷ 100,000
= $0.12

Depreciation expense = Depreciable cost per unit × output units
= $0.12 × 10,000
= $1200

Barb's Florist

	Computation			End of year	
year	units of Activity ×	Depreciable Cost per Unit =	Annual Depreciation expense	Accumulated Depreciation	Book value
2020	10,000 ×	$0.12 =	$1200	$1200	$11800
2021	30,000	0.12	3600	4800	8200
2022	20,000	0.12	2400	7200	5800
2023	25000	0.12	3000	10200	2800
2024	10000	0.12	1200	11400	1600
2025	5000	0.12	600	12000	1000

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(c)

We know:—

Depreciation Expense = Beg NBV × D.B Rate

D.B Rate = Declinning Balance Rate = 2×20% = 40%

Beg Note Book Value = $ 13000

Barb's Florists

Computation

Year	Book Value Beg of year	× D.B Rate	= Annual Depreciation Expense	× Partial year	= Current year Expense	End of year Accumulated Depreciation	End of year Book value
2020	$13000	× 40%	=$5200	× $\frac{9}{12}$	=$3900	$3900	$9100
2021	9100 ×	40%	= 3640		3640	7540	5460
2022	5460	40%	2184		2184	9724	3276
2023	3276	40%	1310		1310	11034	1966
2024	1966	40%	786		786	11820	1180
2025	1180	40%	472 plug		180	12000	1000

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Revising Periodic Depreciation

1. Determine new depreciable cost
2. Divide by remaning useful life.

#Problem 3:

➔Barb's Florists purchased a small delivery truck at a **cost of $13,000** on **January 1, 2020**. DuPage expects the truck to have a **salvage value of $1,000** at the end of its **5-year useful life.**

Barb's Florists decides on **January 1, 2023**, to extend the useful life of the truck **one year** (a total life of six years) and **increase its salvage value to $2,200**. The company has used the straight-line method to depreciate the asset to date. Accumulated depreciation after three years (2020–2022) is $7,200, and book value is $5,800.

a) Calculate the depreciation expense for 2023 and future years?

(a)

Book value, 1/1/23	$5800
Less: Salvage value	2200
Dipreciable cost	$3600
Remaining useful life	3 years (2023-2025)
Revised annual depreciation ($3600 ÷ 3)	$1200

#Problem 4:

Apex company purchased a piece **of equipment for $36,000**. It estimated a **6-year life** and **$6,000 salvage value**. Thus, **straight-line depreciation was $5,000 per year** [($36,000 − $6,000) ÷ 6]. At the end of year three (**before the depreciation adjustment**), it estimated the **new total life to be 10 years** and the new **salvage value to be $2,000.**

a) Compute the revised depreciation.

a

We know:—

Depreciation expense = [Cost − Salvage value]/useful life
= ($36000 − $6000)/ 6
= $5000

∴ Accumulated Depreciation After 2 years = 2 × $5000 = $10,000

∴ Book Value = Cost − Accumulated Depreciation
= $36000 − $10,000
= $26000

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Book value, After 2years of Depreciation	$26000
Less : New Salvage value	2000
Depreciable cost	$24000
Remaning useful life	8 years
Revised Accumulated Depreciation ($24000/8)	$3000

Sale of plant Asset :

=> Compare the book value of the asset with the proceeds recevied form the sale.

1. If proceeds exceed the book value, A Gain on dispo sal occurs.

2. If proceeds are less than the book value, A Loss on disposal occurs.

#Problem 5:

➔Overland Trucking has decided to sell an old truck that cost $30,000 and which has accumulated depreciation of $16,000.

(a) What entry would Overland Trucking make to record the sale of the truck for **$17,000 cash**?

(b) What entry would Overland Trucking make to record the sale of the truck for **$10,000 cash**?

(a)

Journal enterey is:—

	Dr	Cr
Cash	17000	
Accumulated Depreciation	16000	
Equipment		30000
Gain → Just like revenue		3000

(b)

Journal enterey is:—

	Dr	Cr
Cash	10000	
Accumulated Depreciation	16000	
Loss → Just like expense	4000	
Equipment		30000

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#Problem 6:

➔ Barb's Florists purchased a small delivery truck at a cost of $13,000 on April 1, 2020. DuPage expects the truck to have a salvage value of $1,000 at the end of its 5-year useful life.

(a) what will be follows After 5 years of use the balance on your accountants.

(b) What would be the journal entry when we sell the truck after 5 years?

(c) if sell the truck $3,000 so What would be the journal entry?
(d) if sell the truck $300 so What would be the journal entry?

(e) if you sell the car after 2 years, and your salvage value $3,000 so What would be the journal entry?

(a)

⇒ After 5 years of use the balance in your accountants will be as follows:—

Vehicle	13,000	debit balance
Accumulated Depreciation	12,000	credit balance.

(b)

⇒ Journal entry when we sell the truck After 5 year

	Dr	Cr
Cash	1000	
Accumulated Depreciation	12000	
Vehicle		13000

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(c)

⇒ If sell the car $3000 so the journal Entry is:—

	Dr	Cr
Cash	3000	
Accumulated Depreciation	12000	
Vehicle		13000
Gain ↓ Just like Revenue.		2000

(d)

⇒ If sell the truck $300 so the journal Entry is:—

	Dr	Cr
Cash	300	
Accumulated Depreciation	12000	
Loss → Just like expense	700	
Vehicle		13000

(e)

⇒ If you sell the truck After 2years and your salvage value $3000 so. journal entry is

	Dr	Cr
Cash	3000	
Accumulated Depreciation	4800	
Loss	5200	
Vehicle		13000

Inventories

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⇒ chapter 6 in the backup chapter of chapter 3

⇒ The task we did in chapter three.

Ex: (1) purchased $1000 Equipment.

(2) Annual depreciation in $4900 on the building

So, What would be the journal entry.

	Dr	Cr
Equipment	1000	
Cash		1000
Depreciation expense	350	
Accumulated Depreciation - Buildings		350

Now in chapter - 6:—

(3) Let you purchased 60,000 taka of rice. So the Journal Entry is → This is just for trading business

	Dr	Cr
Inventory	60000	
Cash		60000

Again:— (3) Let you purchased 60,000 taka of rice the amount of which is 1200 kg and you sell the $\frac{1}{2}$ of this 40000 tk. So the journal Entry is

	Dr	Cr
Inventory	60000	
Cash		60000

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Cash 40000

Sales Revenue 40000

Now because we sold 50% the rice we have to transfer 50% of the inventory account balance to an expense Account.

So the journal entry is:-

	Dr	Cr
Cost of good sold (COGS)	30,000	
Inventory		30,000

At the end of the month,
you found after counting that you have 600kg of rice in your store.

1200 kg - 600kg = The quantity of sold 600kg

Now we have find the cost of this rice which is called COGS:-

∴ 600kg × 50tk = 30,000 tk that will be our COGS Account.

Now what will happen if you purched 1200 kg at diffrent prices.

Let, 500kg = 40 TK
600kg = 50 TK
100kg = 55 TK

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A list of Inventory and COGS And Counting method

Asset Account → Inventory	Expense Account COGS	Method Counting Formula
Supplies	Supplies Expense	Counting
prepaid Insurance Expence	Insurance Expense	time pass
Building/Equipment/ Vehicle	Depreciation Expense	(Formula)

Difference between Service business and trading business

Service Business	Trading Business
No Inventory Account on Balance sheet	Inventory Account on balance sheet.
Service Revenue	Sales Revenue.

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#Inventory Account Calculation

(1.) 1st we have to count unsold Inventory.

(2.) After count unsold Inventory we called it ending Inventory.

Note:—

=> কোনো কোনো Accountants Inventory Account কে Debit না করে purchased Account এ Debit করে। এবং বছর শেষে purchased Account থেকে Inventory Account এ switch করে।

=> Let, you purchased 50,000 tk of rice the ammount of which is 1000 kg, you sell the half of this 30,000 tk. What should the journal Entry

	Dr	Cr
Inventory	50000	
Cash		50000
Cash	30000	
Sales Revenue		30000

Therefore 25000 taka must be transferred from Inventory to an expense Account

	Dr	Cr
COGS	25000	
Inventory		25000

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In that case:—

» Sales happen every day. Therefore, ideally, we should transfer the Cost(=purchased price) From inventory to expense Account(COGS) Every time we are making sales.

Its possible only if

① Good Real time Accounting software

② Showroom of cars or showrooms of jewellery just like that.

this is very difficult for other businesses to do, so they follows periodic inventory method

Periodic Inventory Method:—

Let, you purchased 50,000 tk of rice the ammount of which is 1000kg. you sell some of this 30000 tk, on Nov 5 so, the journal Entry is:—

		Dr	Cr
NOV 05	purchase	50,000	
	cash		50000

Therefore we must to do Adjusting journals at month end

First AdJE

		Dr	Cr
NOV 30	Inventory	50000	
	purchase		50000

To transfer balance from purchase Account to an inventory Account.

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Second ADJE:-

		Dr	Cr
NOV 30	COGS	25000	
	Inventory		25000

For 2nd AJE, 1st we have to go to a ledger Account of Inventory then Count Unsold good in your store

Inventory

NOV 1 Balance 101,000	
NOV 30 AJE#1 501,000	
NOV 30 Balance 60,000	
	NOV 30 AdJE#2 25000
NOV 30 Balance 35,000	

⇒ 1st Count all unsold good in your store.

NOV 30 Count 700 kg

⇒ 2nd, multiply the counted units with purchased price

700 kg × TK 50 purchased price = TK 35000

= Cost of unsold goods.

= Ending Inventory.

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Sol. Cost of goods sold = 1000 + 50000 − 35000
= 25000 TK

Cost of goods Available for sale

∴ COGS = Beginning Inventory + purchase − Ending Inventory

∴ COGS = \$10000 + \$50000 − \$35000
= \$25000

Cost Flow Assumptions:—

⇒ we have three Alternative Assumptions to solve the problem of which price to multiply with.

① First in First out (FIFO) → multiply with Last price
② Last In First out (LIFO) → multiply with first price.
③ Average cost.

GOOD LUCK

P6.2A **(LO 2)** Glee Distribution markets CDs of the performing artist Unique. At the beginning of October, Glee had in beginning inventory 2,000 of Unique's CDs with a unit cost of $7. During October, Glee made the following purchases of Unique's CDs.

Oct. 3	2,500 @ $8	Oct. 19	3,000 @ $10
Oct. 9	3,500 @ $9	Oct. 25	4,000 @ $11

During October, 10,900 units were sold. Glee uses a periodic inventory system.

Instructions

a. Determine the cost of goods available for sale.

b. Determine (1) the ending inventory and (2) the cost of goods sold under each of the assumed cost flow methods (FIFO, LIFO, and average-cost). Prove the accuracy of the cost of goods sold under the FIFO and LIFO methods.

c. Which cost flow method results in (1) the highest inventory amount for the balance sheet and (2) the highest cost of goods sold for the income statement?

P6.3A **(LO 2)** **Financial Statement** Sekhon Company had a beginning inventory on January 1 of 160 units of Product 4-18-15 at a cost of $20 per unit. During the year, the following purchases were made.

Mar. 15	400 units at $23	Sept. 4	330 units at $26
July 20	250 units at $24	Dec. 2	100 units at $29

1,000 units were sold. Sekhon Company uses a periodic inventory system.

Instructions

a. Determine the cost of goods available for sale.

b. Determine (1) the ending inventory, and (2) the cost of goods sold under each of the assumed cost flow methods (FIFO, LIFO, and average-cost). Prove the accuracy of the cost of goods sold under the FIFO and LIFO methods.

c. Which cost flow method results in (1) the highest inventory amount for the balance sheet, and (2) the highest cost of goods sold for the income statement?

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(a)

Cost of goods sold Available for sale				
Date	Description	Units	Cost per unit	total cost
Oct. 01	Beg. Inventory	2000	$ 7	$ 14000
Oct. 03	purchase	2500	8	20000
Oct. 09	purchase	3500	9	31500
Oct. 19	purchase	3000	10	30000
Oct. 25	purchase	4000	11	44000
	Total	15000		$139500

(b)

Total units	15000	units
Sold units	10900	units
Unsold units	4100	units

FIFO method:—

Ending Inventory [4000×11]+[100×10] = $ 45000

∴ COGS = Beg Inventory + purchase − Ending Inventory

= $ 14000 + $ 125500 − $45000

= $94500

LIFO Method:-

Ending Inventory [2000×7]+[2100×8] = $30800

COGS = $14000 + $125500 − $30800

= $108700

Weighted Average Method:-

Weighted Average cost per unit = total cost/total units

= $130500/15000

= $9.3

∴ Ending Inventory [4100×9.3] = $38130

COGS = $14000 − $12500 − $38130

= $101,370

proof of COGS in FIFO method:-

Date	Description	Units	Cost per unit	total Cost
Oct. 01	Beg. Inventory	2000	$7	$14000
Oct. 03	purchase	2500	8	20000
Oct. 09	purchase	3500	9	31500
Oct. 19	purchase	2900	10	29000
	total	10,900		$94500

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Proof of COGS In LIFO method:—

Date	Discription	Units	Cost per unit	total cost
oct. 25	purchase	4000	$11	$44000
oct. 19	purchase	3000	10	30,000
oct. 09	purchase	3500	9	31500
oct. 03	purchase	400	8	3200
	total	10,900		$108,700

(c)

1. FIFO shows the highest inventory amount for the balance sheet.

2. LIFO shows the highest cost of goods sold (COGS) For the income statement.

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P6.3A8

(a)

Cost of goods Available for sale.				
Date	Discription	Units	Cost per Unit	total Cost
Jan. 01	Beg Inventory	160	$20	$3200
Mar. 15	purchase	400	23	9200
July. 20	purchase	250	24	6000
Sept. 04	purchase	330	26	8580
Dec. 02	purchase	100	29	2900
	total	1240		$29880

(b)

Total units 1240 units
sold units 1000 units

∴ Unsold units 240 units

FIFO Method:-

Ending Inventory $[100\times29]+[140\times26]=\$6540$

$COGS=\$3200+\$26680-\$6540$
$=\$23340$

LIFO Method:-

Ending Inventory $[160\times20]+[80\times23]=\$5040$

$COGS=\$3200+\$26680-\$5040$
$=\$24840$

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Weighted Avarage Method:-

Weighted average cost per unit = $29880 / 1240 units
= $24.097

∴ Ending Inventory [240 units × $24.097] = $5783

COGS = $3200 + $26680 − $5783
= $24097

proof of COGS In FIFO method:-

Date	Discreption	units	cost per unit	total cost
Jan. 01	Beg Inventory	160	$ 20	$3200
Mar. 15	purchase	400	23	9200
July 20	purchase	250	24	6000
Sept. 04	purchase	190	26	4940
	total	1000		$23340

Proof of COGS In LIFO method:-

Date	Description	Units	Cost/ Unit	total cost
Dec. 02	purchase	100	$20	$2000
Sep. 04	purchase	330	26	8580
July 20	purchase	250	24	6000
Mar 15	purchase	320	23	7360
	total	1000		$24840

c

1. FIFO shows the highest inventory ammount for the balance sheet.
2. LIFO shows the highest cost of goods sold for the income statement.

Ch: 10

Managerial Accounting

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Difference between Financial Accounting & Managerial Accounting

	Financial Accounting	Managerial Accounting
purpose of reports	General purpose	used internally by managers only
Number of reports	Four reports	No limits, whatever required
Compulsion	Mandatory once a year	Reports are not mandatory
Orientation	Historical reports	prepared for future
verification	Independent audit	usually not audited

Service Business Vs Trading Business Vs Manufacturing Business

Service	Trading	Manufacturing
⇒ No. Inventory	1. Inventory	1. Raw Materials Inventory
		2. Work-In-process "
		3. Finished Goods "

Main Math in Ch-19

⇒ Make a Cost report of a manufacturing Company.

Manufacturing Business

⇒ There are three types of Inventory Account

1. Raw Materials ⟶ Ware house.
2. Work-In-process ⟶ Factory
3. Finished Goods ⟶ Show Room.

What we Saw in Inventory Account In Chapter 6:-

Dr	Inventory	Cr
April 1 Balance 3,000		
Apr 30 ADJ#1 5000	April 30. ADJ#2	6,000
Apr 30 Balance 2,000		

∴ COGS = Beg. Inventory + purchase − Ending Inventory
= 3000 + 5000 − 2000
= 6000

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Normal Journal:—

		Dr	Cr
Apr 15	purchase	5,000	
	Cash		5000

Note:— factory related যে কোন ধরনের expense — হবে আমাদের Asset

Adjusting Journal:—

		Dr	Cr
Apr 30,	Inventory	5000	
	purchase		5000
Apr 30,	COGS	6000	
	Inventory		6000

=> This same pattern will be followed in the three Inventory Accounts of a manufacturing Company.

Raw Materials Inventory Account:—

Dr — Raw-Materials — Cr

Dr		Cr	
April 1 Balance	4,000		
April 15 Cash	9,000	Apr 30 Direct Material	10,000
April 30 Balance	3000		

Direct Material = Beg. Raw Material + Raw Material purchase − Ending Raw Material

= 4000 + 9000 − 3000

= 10,000

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Normal Journal:—

			Dr	Cr
April 15	Raw Material		0000	
		Cash		0000

ADJ/ Adjusting Journal:—

			Dr	Cr
April 30	Work-In-process		10,000	
		Raw Material		10,000

Work-In-process Inventory Account:—

W-I-P

Dr				Cr
April 1 Balance	5000			
April 30 Direct Material	10,000			
April 30, Direct Labour	8,000			
April 30 M.O.H	3,000	Apr 30	COGM	20,000
April 30. Balance	6000			

$$COGM = Beg\ WIP + D.M + D.L + MOH - Ending\ WIP$$
$$= 5000 + 10000 + 8000 - 6000$$
$$= 20,000$$

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Note:-

⇒ All expenses which are related to manufacturing must be debited to WIP instead of expense accounts. Indirect labour and Indirect material are also part of the manufacturing overhead. (MOH)

Adjusting Journal:-

		Dr	Cr
April 30	Finished Goods	20000	
	WIP		20000

Finished Goods Inventory Account

Dr — Finished Goods — Cr

Dr		Cr	
April 1 Balance	7,000		
April 30 COGM	20,000	April 30 COGS	10000
April 30 Balance	17000		

COGS = Beg. F.G Inventory + COGM − Ending F.G. Inventory

= 7000 + 20,000 − 17000

= 10000

"GOOD LUCK"

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Adjusting Journal

		Dr	Cr
April 30	Cost of Goods Sold	10000	
	F. G Inventory		10000

Now your Owner wants a Cost report

⇒ Go to WIP And Find COGM. that amount in the Cost of goods manufactured during the month.

We know

$$COGM = \text{Beg WIP} + DM + D.L + MOH - \text{Ending WIP}$$

$$= 5000 + 10000 + 8000 + 3000 - 6000$$

$$= 5000 + 4000 + 10000 - 3000 + 8000 + 3000 - 6000$$

$$= 20000$$

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#Cost Report:-

ABC Company
Schedule of Cost of Goods Manufactured.
For Month ended April 30, 2020

Work-In-process-beg		5,000
Direct Material:		
R.M beg	4,000	
'+' R.M bought	9,000	
R.M Avaible	13,000	
'-' Ending. R.M	3,000	10,000
Direct Labour		8,000
M.O.H		3,000
Total Cost W.I.P		26,000
Less:- W.I.P - Ending		6,000
Cost of goods Manufactured		20,000

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Green Company is a manufacturer of computers. Its chief accountant resigned in October 2023. An inexperienced junior accountant has prepared the following income statement for the month of October 2023.

Green Company
Income Statement
For the Month Ended October 31, 2023

Sales (net)		$780,000
Less: Operating expenses		
Raw materials purchased	$264,000	
Wages – factory workers	190,000	
Advertising expense	90,000	
Salaries – office employees	75,000	
Rent on factory facilities	60,000	
Depreciation on sales equipment	45,000	
Depreciation on factory equipment	31,000	
Indirect labor cost	28,000	
Utilities expense	12,000	
Insurance expense	8,000	803,000
Net loss		$23,000

Prior to October 2023, the company had been profitable every month. The company's president is concerned about the accuracy of the income statement. As her friend, you have been asked to review the income statement and make necessary corrections. After examining the data, you have acquired additional information as follows.

1. Inventory balances at the beginning and end of October were:

	October 1	October 31
Raw materials	$18,000	$34,000
Work in process	16,000	14,000
Finished goods	30,000	48,000

2. Only 70% of the utilities expense and 60% of the insurance expense apply to factory operations. The remaining amounts relate to selling and administration.

Instructions

(a) Prepare a schedule of cost of goods manufactured for October 2023.

(b) Prepare a correct income statement for October 2023.

TOPIC NAME : DAY : TIME : DATE : / /

(a)

Green Company

Schedule of Cost of Goods Manufactured

For the Month Ended Octor 31, 2023

Work-In-Process - October 1		$16000
Dereet materials:		
Raw materials Beginning	$18,000	
Raw materials purchased	264,000	
Raw materials Avialable	282000	
Ending Raw materials	34000	
Direct materials used	$248,000	
Dereet Labour	190,000	
Manufacturing Overhead:		
Factory rent	60,000	
Depreciation factory equip.	31000	
Inderect Labour	28000	
Factory utilities (12000×70%)	8400	
Factory insurance (8000×70%)	4800	
Total	132,200	
Total manufacturing costs		570200
Total Cost of Work-in-process		586200
Less: WIP - Ending ~~Oct 31~~		14000
Cost of goods manufacture		$572,200

TOPIC NAME: DAY: TIME: DATE: / /

(b)

Green Company
Income Statement
For the Month Ended October 31, 2023

Sales (net)		$780,000
Less: Cost of Goods sold		
Finished goods - October 1	$30,000	
Cost of goods Manufactured	570080	
Cost of goods available for sale	602,200	
Less: Finished goods - October - 31	48000	
Cost of goods sold		554200
Gross profit		225,800
Less: Operating expenses:-		
Adv. expenses	90,000	
Salaries - office employees	75000	
Depreciation - Sales Equipment	45000	
Utility expense (12000 X 30%)	3600	
Insurance expense (8000 X 40%)	3200	
Total operating expense		216,800
Net Income		$ 9000

Solution:

Green Company
Schedule of Cost of Goods Manufactured
For the Month Ended October 31, 2023

Work in process – October 1			$16,000
Direct materials:			
Raw materials – October 1	$18,000		
Raw materials purchased	264,000		
Raw materials available	282,000		
Raw materials – October 31	34,000		
Direct materials used		$248,000	
Direct labour		190,000	
Manufacturing overhead:			
Factory rent	60,000		
Depreciation factory equip.	31,000		
Indirect labour	28,000		
Factory utilities (12,000X70%)	8,400		
Factory insurance (8,000X60%)	4,800		
Total manufacturing overhead		132,200	
Total manufacturing costs			570,200
Total cost of work in process			586,200
Less: Work in process – October 31			14,000
Cost of goods manufactured			$572,200

Green Company
Income Statement
For the Month Ended October 31, 2023

Sales (net)		$780,000
Less: Cost of goods sold		
Finished goods – October 1	$30,000	
Cost of goods manufactured	572,200	
Cost of goods available for sale	602,200	
Less: Finished goods – October 31	48,000	
Cost of goods sold		554,200
Gross profit / Gross margin		225,800
Less: Operating expenses		
Advertising expense	90,000	
Salaries – office employees	75,000	
Depreciation - sales equipment	45,000	
Utility expense (12,000X30%)	3,600	
Insurance expense (8,000X40%)	3,200	
Total operating expense		216,800
Net income		$9,000

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Merchandising Business

Merchandising business normally has one inventory account

For merchandising business, the cost of goods sold is calculated as follows:

COGS = Beginning Inventory + Purchases – Ending Inventory

Manufacturing Business

Manufacturing business normally has three inventory accounts:

(1) Raw materials (2) Work in process (3) Finished goods

For manufacturing business, the cost of goods sold is calculated as follows:

COGS = Beginning Finished Goods + Cost of Goods Manufactured – Ending Finished Goods

And the cost of goods manufactured is calculated as follows:

COGM = Beginning W-I-P + Direct Material + Direct Labour + Mfg. Overhead – Ending W-I-P

And the direct material is calculated as follows:

DM = Beginning Raw Materials + Purchase of Raw Materials – Ending Raw Materials

Presentation of Income Statement in Merchandising and Manufacturing Businesses

Expenses are subtracted from the Revenues in multi-steps.

First, Cost of Goods Sold is subtracted from Sales Revenue. The difference is called Gross Profit or Gross Margin.

Then, all other operating expenses are subtracted from the Gross Profit / Margin. The difference is called Operating Income.

Then, non-operating expenses are subtracted from the Operating Income. The difference is called Income before taxes.

Finally, income taxes are subtracted, and the result is Net Profit or Net Income

Ch:22

Cost Volume Profit (CVP)

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- Income Statement is a report. Think about it as a report card.
- What does an income statement report to us? Answer: Profit or net income.
- If the owner or manager is happy with the report card, then it is fine.
- If she is not satisfied, she will think about how to increase profit in the next period.
- To increase profit, sales must be increased. Cutting expenses is not enough.
- So, it boils down to the question:
    - How much the sales should be if you want a certain amount of profit?

Example:

ABC Limited
Income Statement
For the month ended Nov 30, 2023

Sales Revenue [3000 unit]		750,000
Advertising Expense	45,000	
Cost of Goods Sold	210,000	
Rent Expense	60,000	
Sales Commission	90,000	
Office Salaries	75,000	
Utility Expense	20,000	
		500,000
Net Profit		250,000

Now, we must find out a general formula which will tell us what the sales should be if we desire a certain amount of profit. This formula will help us to set our future sales target. Please note that COGS and sales commission are variable expenses. The rest are fixed expenses.

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To derive the formula, fixed expenses and variable expenses must be treated differently. Therefore, we will recast the income statement in a different format [CVP format].

ABC Limited
Income Statement
For the month ended Nov 30, 2023

Sales Revenue (3,000 units)	Tk. 750,000
Less: Variable Costs	300,000
Contribution Margin	450,000
Less: Fixed Costs	200,000
Net Profit / Net Income	Tk. 250,000

Then we must find out per unit sales and per unit variable costs because these two items are constant at unit level. Fixed costs are fixed (constant) at total amount but not constant at unit level. So fixed cost per unit is meaningless.

ABC Limited
Income Statement
For the month ended Nov 30, 2023

		Unit	Ratio
Sales Revenue (3,000 units)	Tk. 750,000	Tk. 250	100%
Less: Variable Costs	300,000	100	40%
Contribution Margin	450,000	Tk. 150	60%
Less: Fixed Costs	200,000		
Net Profit / Net Income	Tk. 250,000		

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Therefore, sales quantity can be expressed as:

Sales (in units) = [Fixed Cost + Profit] / CM per Unit (1)

And if we multiply both sides by selling price per unit, then we get:

Sales (in Taka) = [Fixed Cost + Profit] / CM Ratio (2)

We will use these two formulas to solve the problems in chapter 22.
Derivation of the formula most likely will **not** be tested in the exam.

Additional formulas in chapter 22:

Margin of Safety = Actual Sales - Break Even Sales

MS Percentage = Margin of Safety / Actual Sales X 100%

Degree of Operating Leverage = Contribution Margin / Net Profit

Break Even Sales is a level of sales where there is no profit or no loss.

Note: See Mega Manufacturing Limited question with solution uploaded in Teams.

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Question: Mega Manufacturing Limited

The income statement of Mega Manufacturing Limited in contribution margin format is given below:

Mega Manufacturing Limited
Income Statement
For the month ended March 31, 2023

Sales (12,000 units)	Tk.	24,00,000
Less: Variable Costs		18,00,000
Contribution Margin		6,00,000
Less: Fixed Costs		2,40,000
Net Profit	Tk.	3,60,000

Required:

(a) Find contribution margin per unit.
(b) Find contribution margin ratio.
(c) How much is break-even sales in units?
(d) How much is break-even sales in taka?
(e) If company wants Tk. 4,80,000 profit next month, then how much should be sales in units?
(f) If company wants Tk. 4,80,000 profit next month, then how much should be sales in taka?
(g) Calculate margin of safety in taka and in percentage
(h) Calculate degree of operating leverage

Solution:

(a) Contribution margin per unit = Contribution margin / Units sold
= Tk 6,00,000 / 12,000 units
= Tk. 50 per unit

(b) Contribution margin ratio = Contribution margin / Sales
= Tk. 6,00,000 / Tk. 24,00,000
= 0.25 or 25%

(c) Break even sales (in units) = (Fixed Cost + Profit) / CM per unit
= (Tk. 2,40,000+ 0)/Tk. 50
= 4,800 units

(d) Break even sales (in taka) = (Fixed Cost + Profit) / CM ratio
= (Tk. 2,40,000+0)/0.25
= Tk. 9,60,000

(e) Desired target sales (in units) = (Fixed Cost + Profit) / CM per unit
= (Tk. 2,40,000+Tk. 4,80,000)/Tk. 50
= 14,400 units

(f) Desired target sales (in taka) = (Fixed Cost + Profit) / CM ratio
= (Tk. 2,40,000+Tk. 4,80,000) / 0.25
= Tk. 28,80,000

(g) Margin of safety = Actual sales - Break even sales
= Tk. 24,00,000 - Tk. 9,60,000
= Tk. 14,40,000

(g) Margin of safety percentage = Margin of safety / Actual sales
= Tk. 14,40,000 / Tk. 24,00,000
= 0.6 or 60%

(h)	Degree of operating leverage	=	Contribution margin / Net Profit
		=	Tk. 6,00,000 / Tk. 3,60,000
		=	1.67 times

Formulas:

Contribution margin per unit	=	Contribution margin / Units sold
Contribution margin ratio	=	Contribution margin / Sales
Sales (in units)	=	(Fixed Cost + Profit) / CM per unit
Sales (in dollars)	=	(Fixed Cost + Profit) / CM ratio
Margin of safety	=	Actual sales - Break even sales
Margin of safety percentage	=	Margin of safety / Actual sales
Degree of operating leverage	=	Contribution margin / Operating profit

Break Even Sales is the level of sales when there is no profit or no loss.
Therefore, at break even level the profit is zero.

Example: A salesgirl in a saree shop sold the following number of sarees and earned her salaries in the first six month of the year. Her salary includes a fixed amount plus commission based on number of sarees sold by her.

	Sarees Sold	Total Salary (including commission)
January	20,000	Tk. 30,000
February	40,000	48,000
March	35,000	49,000
April	50,000	63,000
May	30,000	42,000
June	43,000	61,000

Required:
(a) Find her fixed salary and commission per saree using high-low method.
(b) Express her total salary in an equation format.
(c) If she sells 45,000 sarees next month, what will be her total salary?

Solution:

(a)

	Sarees Sold	Total Salary	
Highest month	50,000	Tk. 63,000	F + V
Lowest month	20,000	30,000	F + V
Difference	30,000	Tk. 33,000	diff. V

Therefore, commission per saree [variable cost] will be = Tk. 33,000 / 30,000 sarees
= Tk. 1.10 per saree.

Her fixed salary, using highest month:
Her total commission in highest month = 50,000 sarees X Tk. 1.10 = Tk. 55,000
Therefore, her fixed salary was Tk. 63,000 less Tk. 55,000 = Tk. 8,000

We can also find her fixed salary, using lowest month. We will get the same result.
Her total commission in lowest month = 20,000 sarees X Tk. 1.10 = Tk. 22,000
Therefore, her fixed salary was Tk. 30,000 less Tk. 22,000 = Tk. 8,000

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(b) We can express her total salary with the following equation:

$$Y = \text{Tk. } 8{,}000 + \text{Tk. } 1.10X$$

Where: Y ➔ Total salary

X ➔ Number of sarees sold

Tk. 8,000 ➔ Fixed salary per month.

Tk. 1.10 ➔ Variable commission per saree sold.

(c) Next month if she can sell 45,000 sarees, then her total salary will be:

$$Y = \text{Tk. } 8{,}000 + \text{Tk. } 1.10X$$

$$\Rightarrow Y = \text{Tk. } 8{,}000 + \text{Tk.} 1.10 * 45{,}000$$

$$\Rightarrow Y = \text{Tk. } 57{,}500 \quad \text{(Answer).}$$